Theory of Elementary Particles
Lecture 11

Krzysztof M. Graczyk
kgraczyk.github.io

Institute of Theoretical Physics
University of Wrocław
pl. M. Borna 9, 50-204 Wrocław, Poland

15.12.2026, Wrocław



Vector Fields
- Spin 1 Vector Field
- Massless vector field: spin 1

Higgs Mechanism
- Abelian Case: U(1) gauge symmetry
  - Standard Derivation
- General remarks
- Effective photon mass from symmetry breaking
- Unitary gauge
  - Abelian Case
  - Non-Abelian Case
- 't Hooft gauge

Spin 1 Vector Field

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + \frac{1}{2} M^2 A_\mu A^\mu, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{1}$$

Euler-Lagrange equations:

$$\partial_\mu F^{\mu\nu} + M^2 A^\nu = 0 \tag{2}$$

then

$$\Box A^\nu - \partial^\nu \partial_\mu A^\mu + M^2 A^\nu = 0 \tag{3}$$

$$\partial_\nu \left(\Box A^\nu - \partial^\nu \partial_\mu A^\mu + M^2 A^\nu \right) = 0 \quad \Rightarrow \quad \partial_\nu A^\nu = 0. \tag{4}$$

Hence

$$\Box A^\nu = -M^2 A^\nu \quad \Rightarrow \quad A^\nu \sim \varepsilon^\nu(q) e^{iq\cdot x} \quad \Rightarrow (q^2 - M^2)\varepsilon^\mu(q) = 0 \tag{5}$$

and

$$\varepsilon^\nu q_\nu = 0 \tag{6}$$

In the rest frame of the particle

$$\boldsymbol{\varepsilon}(\lambda = +1) = -\frac{1}{\sqrt{2}}(1, i, 0) \tag{7}$$

$$\boldsymbol{\varepsilon}(\lambda = 0) = (0, 0, 1) \tag{8}$$

$$\boldsymbol{\varepsilon}(\lambda = -1) = \frac{1}{\sqrt{2}}(1, -i, 0) \tag{9}$$

In the laboratory frame

$$\varepsilon^{\mu}(\lambda, \mathbf{q}) = (\varepsilon^{0}(\lambda, \mathbf{q}), \boldsymbol{\varepsilon}(\lambda, \mathbf{q})) \tag{10}$$

where

$$\varepsilon^{\mu}(\lambda = 0, \mathbf{q}) = \frac{1}{M}(|\mathbf{q}|, 0, 0, E) \tag{11}$$

and

$$\varepsilon^{\mu}(\lambda = \pm 1, \mathbf{q} = |\mathbf{q}|\hat{\mathbf{z}}) = (0, \boldsymbol{\varepsilon}(\lambda = \pm 1)) \tag{12}$$

where

$$\sum_{\lambda = \pm 1, 0} \varepsilon^{\mu *}(\lambda)\varepsilon^{\nu}(\lambda) = -g^{\mu\nu} + \frac{q^{\mu}q^{\nu}}{M^2} \tag{13}$$

Massless vector field: spin 1

Massless Vector Field

$$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu}, \quad F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{14}$$

Euler-Lagrange equations:

$$\partial_\mu F^{\mu\nu} = 0 \tag{15}$$

which leads to

$$\Box A^\nu - \partial^\nu \partial_\mu A^\mu = 0 \tag{16}$$

Suppose we work in the Lorenz gauge $\partial_\mu A^\mu = 0$. Hence

$$\Box A^\nu = 0 \tag{17}$$

Then

$$\Rightarrow \quad A^\nu \sim \varepsilon^\nu(q) e^{iq\cdot x} \quad \Rightarrow q^2 \varepsilon^\mu(q) = 0 \quad \Rightarrow q^2 = 0 \tag{18}$$

and

$$\varepsilon^\nu q_\nu = 0 \tag{19}$$

- $$q_0^2 = \mathbf{q}^2 \tag{20}$$

  and

$$\varepsilon_0 = \frac{\boldsymbol{\varepsilon}\cdot\mathbf{q}}{q_0} = \frac{\boldsymbol{\varepsilon}\cdot\mathbf{q}}{|\mathbf{q}|} = \varepsilon_{\parallel} \tag{21}$$

- On the other hand one can use gauge invariance and redefine photon field

$$A^\mu(x) \to A^\mu(x) - \partial^\mu\Lambda(x) \Rightarrow \varepsilon^\mu(q) \to \varepsilon^\mu(q) - q^\mu\Lambda(q) \tag{22}$$

- Suppose we choose $\Lambda(q) = \frac{\boldsymbol{\varepsilon}\cdot\mathbf{q}}{q_0^2}$.
- Then the photon polarization after gauge transformation is completely transverse to the photon momentum, namely,

$$\varepsilon^\mu = (0, \boldsymbol{\varepsilon}_\perp), \quad \boldsymbol{\varepsilon}_\perp = \boldsymbol{\varepsilon} - \mathbf{q}\frac{\boldsymbol{\varepsilon}\cdot\mathbf{q}}{|\mathbf{q}|^2} = \boldsymbol{\varepsilon} - \boldsymbol{\varepsilon}_{\parallel} \tag{23}$$

- Vector $\boldsymbol{\varepsilon}_\perp$ is a combination of the two orthogonal transverse polarizations!
- * **Hence massless vector field has two degrees of freedom!**

Higgs Mechanism

Abelian Case: U(1) gauge symmetry

$$\mathcal{L}_{U(1)} = (D_\mu \phi)^\dagger (D^\mu \phi) - V(\phi) - \frac{1}{4} F_{\mu\nu} F^{\mu\nu} \tag{24}$$

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2 \tag{25}$$

where

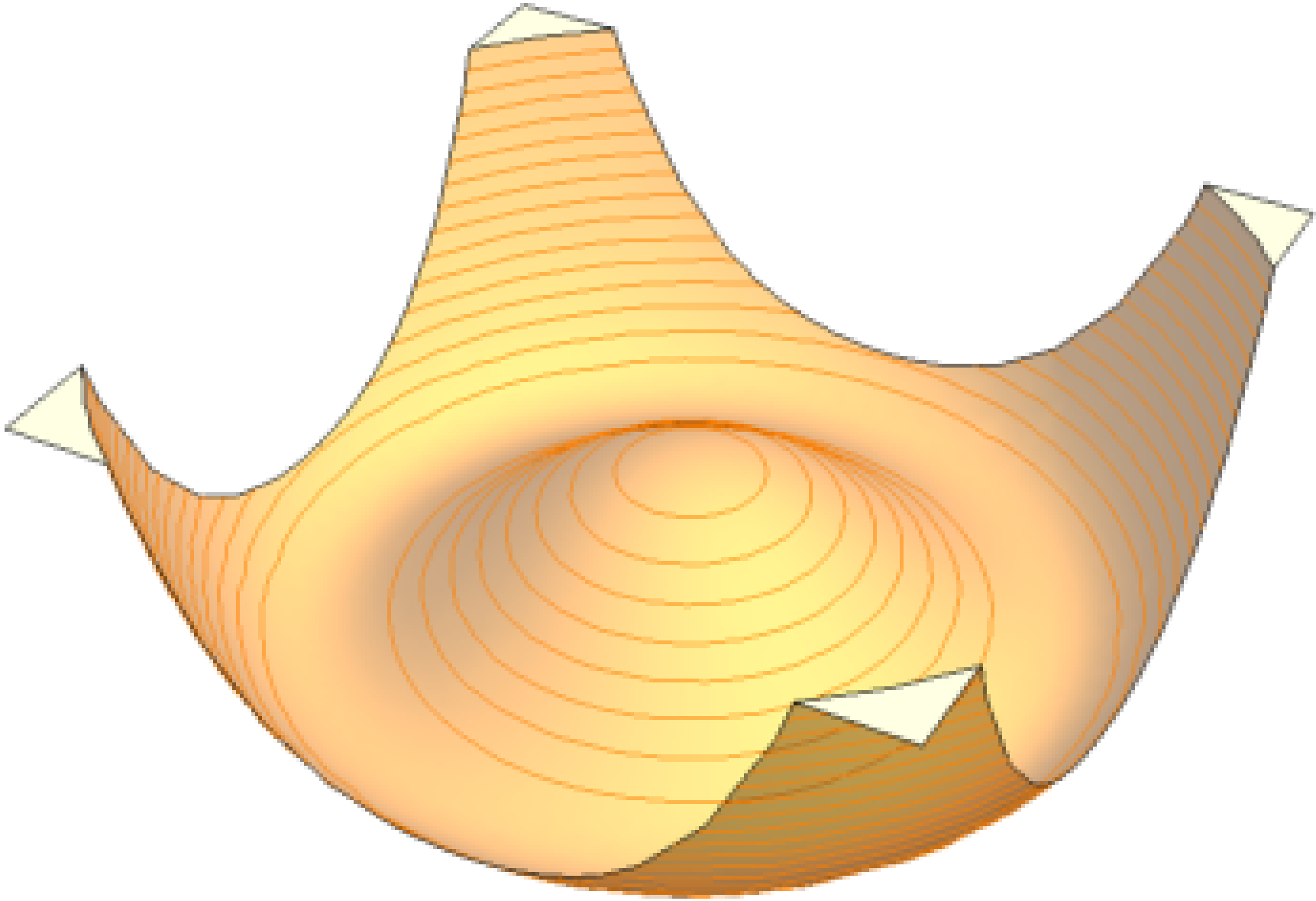
$$\phi = \frac{1}{\sqrt{2}} (\phi_1 + i\phi_2) \tag{26}$$

and the covariant derivative reads

$$D_\mu = \partial_\mu - igA_\mu \tag{27}$$

$U(1)$ local transformations:

- $\phi(x) \to \phi'(x) = e^{-i\alpha(x)} \phi(x)$
- $A_\mu \to A'_\mu = A_\mu - (1/g)\partial_\mu \alpha(x)$



Standard Derivation

- minimum condition

$$0 = \frac{\partial V(\phi)}{\partial(\phi^\dagger \phi)} = -\mu^2 + 2\lambda(\phi^\dagger \phi)$$

The minimum exists only if $\mu^2 \lambda > 0$.

$$|\phi| = \frac{v}{\sqrt{2}}, \quad v = \sqrt{\frac{\mu^2}{\lambda}} \tag{28}$$

where

$$\phi = \frac{1}{\sqrt{2}}(\phi_1 + i\phi_2) \tag{29}$$

- The symmetry is broken by choosing the vacuum configuration i.e.

$$\langle 0 \mid \phi_1 \mid 0 \rangle = v \quad \text{and} \quad \langle 0 \mid \phi_2 \mid 0 \rangle = 0$$

- Let us redefine the fields

$$\phi'_1 = \phi_1 - v \quad \text{and} \quad \phi'_2 = \phi_2 \tag{30}$$

- then

$$\begin{aligned}
* &= (D_\mu \phi)^\dagger (D^\mu \phi) && (31)\\
&= \frac{1}{2}(\partial_\mu + igA_\mu)\phi^\dagger(\partial^\mu - igA^\mu)\phi && (32)\\
&= \frac{1}{2}(\partial_\mu + igA_\mu)(\phi_1 - i\phi_2)(\partial^\mu - igA^\mu)(\phi_1 + i\phi_2) \\
&= \frac{1}{2}\left[\partial_\mu\phi_1\partial^\mu\phi_1 + \partial_\mu\phi_2\partial^\mu\phi_2 + g^2 A_\mu A^\mu(\phi_1^2 + \phi_2^2)\right] \\
&\quad -gA^\mu(\phi_1\partial_\mu\phi_2 - \phi_2\partial_\mu\phi_1) && (33)\\
\rightarrow &\frac{1}{2}\left[\partial_\mu\phi'_1\partial^\mu\phi'_1 + \partial_\mu\phi'_2\partial^\mu\phi'_2 + g^2 A_\mu A^\mu(\phi'^2_1 + v^2 + 2v\phi'_1 + \phi'^2_2)\right] \\
&\quad -gA^\mu(\phi'_1\partial_\mu\phi'_2 + v\partial_\mu\phi'_2 - \phi'_2\partial_\mu\phi'_1) && (34)
\end{aligned}$$

- a term

$$\frac{g^2 v^2}{2} A_\mu A^\mu \tag{35}$$

  is interpreted as a mass term of gauge boson A^μ, $M_A^2 = v^2 g^2$
- a term $A^\mu \partial_\mu \phi_2$ describes conversion of vector field to scalar...

Potential term

$$
\begin{aligned}
V(\phi) \quad &\rightarrow \quad -\frac{\mu^2}{2}((\phi'_1+v)^2+\phi'^2_2)+\frac{\lambda}{4}((\phi'_1+v)^2+\phi'^2_2)^2 && (36)\\
&= \quad -\frac{\mu^2}{2}(\phi'^2_1+v^2+2v\phi'_1+\phi'^2_2)+\frac{\lambda}{4}((\phi'_1+v)^2+\phi'^2_2)^2 && (37)\\
&= \quad -\frac{\mu^2}{2}(\phi'^2_1+v^2+2v\phi'_1+\phi'^2_2)+\frac{\lambda}{4}(\phi'^2_1+v^2+2v\phi'_1+\phi'^2_2)^2 && (38)\\
&= \quad -\frac{\mu^2}{2}(\phi'^2_1+\phi'^2_2)-\frac{\mu^2}{2}v^2-2\frac{\mu^2}{2}v\phi'_1 \\
&\quad +\frac{\lambda}{4}(\phi'^2_1+\phi'^2_2)^2+\frac{\lambda}{4}(v^4+4v^2\phi'^2_1+4v^3\phi'_1)+\frac{\lambda}{2}v^2(\phi'^2_1+\phi'^2_2) \\
&\quad +\lambda v\phi'_1(\phi'^2_1+\phi'^2_2) && (39)\\
&\rightarrow \quad \lambda=\frac{\mu^2}{v^2} && (40)\\
&= \quad \mu^2\phi'^2_1+\frac{\mu^2}{4v^2}\phi'^4_1+\frac{\mu^2}{2v^2}\phi'^2_1\phi'^2_2+\frac{\mu^2}{4v^2}\phi'^4_2+\frac{\mu^2}{v}\phi'^3_1+\frac{\mu^2}{v}\phi'^2_2\phi'_1-\frac{\mu^2v^2}{4} \\
& && (41)
\end{aligned}
$$

Finally the full Lagrangian (after breaking) reads

$$
\begin{aligned}
\mathcal{L} \quad = \quad & \frac{1}{2}\left[\partial_\mu \phi'_1 \partial^\mu \phi'_1 + \partial_\mu \phi'_2 \partial^\mu \phi'_2 + g^2 A_\mu A^\mu (\phi'^2_1 + v^2 + 2v\phi'_1 + \phi'^2_2)\right] \\
& -gA^\mu(\phi'_1 \partial_\mu \phi'_2 + v\partial_\mu \phi'_2 - \phi'_2 \partial_\mu \phi'_1) \\
& -\left(\mu^2 \phi'^2_1 + \frac{\mu^2}{4v^2}\phi'^4_1 + \frac{\mu^2}{2v^2}\phi'^2_1\phi'^2_2 + \frac{\mu^2}{4v^2}\phi'^4_2 + \frac{\mu^2}{v}\phi'^3_1 + \frac{\mu^2}{v}\phi'^2_2\phi'_1 - \frac{\mu^2 v^2}{4}\right) \\
& -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} \qquad (42) \\
= \quad & \frac{1}{2}\left[\partial_\mu \phi'_1 \partial^\mu \phi'_1 - 2\mu^2 \phi'^2_1\right] + \frac{1}{2}\partial_\mu \phi'_2 \partial^\mu \phi'_2 + \frac{g^2 v^2}{2} A_\mu A^\mu - \frac{1}{4}F_{\mu\nu}F^{\mu\nu} \\
& +\frac{1}{2}\left[g^2 A_\mu A^\mu (\phi'^2_1 + 2v\phi'_1 + \phi'^2_2)\right] - gA^\mu(\phi'_1 \partial_\mu \phi'_2 + v\partial_\mu \phi'_2 - \phi'_2 \partial_\mu \phi'_1) \\
& -\mu^2\left(\frac{\phi'^4_1}{4v^2} + \frac{1}{2v^2}\phi'^2_1\phi'^2_2 + \frac{\phi'^4_2}{4v^2} + \frac{\phi'^3_1}{v} + \frac{\phi'^2_2\phi'_1}{v} - \frac{v^2}{4}\right) \qquad (43)
\end{aligned}
$$

Higgs field: $M_H = \sqrt{2}\mu$, **would-be-Goldstone with mass** 0, **Massive vector field:** $M_A = gv$

Remarks

- The requirement of gauge invariance is the fundamental clue for constructing the renormalized theory.
- In QED the mass term is not gauge invariant!
- Weak interactions $\leftrightarrow$ short-range interactions, hence, the intermediate boson must be of large mass!!!

$\rightarrow$ For renormalizability of the theory one needs to have gauge invariant theory with massive fields!

General remarks

Nambu 1960

Physical vacuum of quantum field theory is analogous to the ground state of the interacting many-body system - it is simply the state of minimum energy, the stable (or equilibrium) configuration.

The situation, in which the ground-state configuration does not display the symmetry of the Hamiltonian is frequently described by saying that the symmetry is spontaneously broken.

Let

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2 \tag{44}$$

for the ground state we have

$$|\phi|^2 = \frac{v^2}{2} \tag{45}$$

hence

$$\langle \phi \rangle_0 = \frac{v}{\sqrt{2}} \tag{46}$$

but only one point in the circle can be chosen for the ground state!

Remark

Different choice for the ground state should lead to the same physics!

Effective photon mass from symmetry breaking

Let us consider the massless QED

- $$\mathcal{L} = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} = -\frac{1}{2} \partial_\mu A_\nu F^{\mu\nu} \tag{47}$$
- E-L equations:
$$\partial_\mu F^{\mu\nu} = 0 \tag{48}$$
then we get
$$\Box A_\nu - \partial_\nu \partial^\mu A_\mu = 0 \tag{49}$$
In Lorentz gauge we have
$$\Box A_\nu = 0 \tag{50}$$
- In the case of the interaction E-M reads
$$\Box A_\nu = j_\nu \tag{51}$$

- Consider complex scalar field with $U(1)$ symmetry, the conserved current reads

$$j^\mu = iq\left(\phi^*\partial^\mu\phi - \partial^\mu\phi^*\phi\right) \tag{52}$$

- This current is not gauge invariant, to have gauge invariance the covariant derivative is introduced

$$\partial^\mu \to D^\mu = \partial^\mu + iqA^\mu \tag{53}$$

(sign convention: the scalar here carries the opposite charge to the Higgs-mechanism subsection, $q \leftrightarrow -g$)

-

$$\begin{aligned} j^\mu &= iq\left(\phi^* D^\mu\phi - D^\mu\phi^*\phi\right) && (54)\\ &= iq\left(\phi^*\partial^\mu\phi - \partial^\mu\phi^*\phi\right) - 2q^2|\phi|^2A^\mu && (55)\end{aligned}$$

- Now after symmetry breaking

$$|\phi| = \frac{v}{\sqrt{2}}$$

we get

$$\Box A^\nu \to (...) - q^2v^2A^\nu \tag{56}$$

so photon gets the mass $m_A = qv$!

Gauge quanta become massive when propagating in the Higgs field — a scalar background field!

Remark

Once a quantum field with a non-zero equilibrium value in the ground state is postulated, its value will in general have a non-zero phase.

Remark

If the field carries a charge, a change in its phase can be compensated by gauge transformation on the gauge field A^{μ}

Let's keep the phase of the vacuum in the following form

$$\langle\phi\rangle_0 = \frac{v}{\sqrt{2}}\exp(i\theta(x)) \tag{57}$$

Then

$$\begin{aligned}
\langle j^\mu(\phi)\rangle_0 &= iq\left(\phi^*\partial^\mu\phi - \partial^\mu\phi^*\phi\right) - 2q^2|\phi|^2A^\mu && (58)\\
&= \frac{qv^2}{2}\left(-\partial^\mu\theta - \partial^\mu\theta\right) - q^2v^2A^\mu && (59)\\
&= -q^2v^2\left(A^\mu + \frac{1}{q}\partial^\mu\theta\right) && (60)
\end{aligned}$$

Then the equation of motions reads

$$\Box A^\nu - \partial^\nu\partial_\mu A^\mu = -q^2v^2\left(A^\nu + \frac{1}{q}\partial^\nu\theta\right) \tag{61}$$

But the gauge transformations has the form

$$A'^{\mu} = A^{\mu} + \frac{1}{q}\partial^{\mu}\chi \tag{62}$$

where $\chi(x)$ is an arbitrary function. Hence we can choose $\chi = \theta$ and redefine the fields and get

$$\Box A'^{\nu} - \partial^{\nu}\partial_{\mu}A'^{\mu} = -q^2 v^2 A'^{\nu} \tag{63}$$

where $M_A = qv$ is the mass of A_{μ} gauge field!

Then if one takes

$$0 = \partial_\nu(\Box A'^\nu - \partial^\nu \partial_\mu A'^\mu) = -q^2 v^2 \partial_\nu A'^\nu \tag{64}$$

so

$$\partial_\nu A'^\nu = 0 \tag{65}$$

Remark

One of the degrees of freedom — the gauge phase — is eaten by the A'_μ field and becomes the longitudinal polarization of the massive gauge boson. This particular choice of the gauge is called unitary!In this gauge the spectrum of masses of particles is physical!

Hidden Symmetry

Remark

If we do not use the gauge freedom in A^μ to reduce the phase to zero, then the gauge choice we made will result in additional factors in the equation of motion. Hence the propagator of the gauge boson will contain the gauge factor!

Remark

Hence one needs to include in the Feynman rules the diagrams coming from the unphysical degrees of freedom!

- In the Higgs mechanism gauge invariance is not lost but **Hidden** and it is encoded in the relations between masses and couplings. This mechanisms is called ***Hidden gauge invariance***.

't Hooft, 1971

Massive vector theories, in which the vector particles acquire mass through the vacuum screening current mechanism are renormalizable.

Llewellyn-Smith 1974

Field theories in which the relations among the parameters of the theory are precisely those required to bring about cancellation among graphs with bad high-energy behavior and render the massive vector boson theory renormalizable.

Unitary gauge

- Let us parametrize the field $\phi(x)$ in polar variables

$$\phi(x) = \frac{1}{\sqrt{2}}(v + \eta(x)) \exp\left[\frac{i\xi(x)}{v}\right] \tag{66}$$

- If one expands around minimum we get

$$\phi(x) \approx \frac{1}{\sqrt{2}}\left(v + \eta(x) + i\xi(x) + ...\right) \tag{67}$$

- Around minimum we have

$$\begin{aligned} \phi^\dagger\phi &= \frac{1}{2}(v + \eta(x))^2 && (68) \\ \partial_\mu\phi^\dagger\partial^\mu\phi &\approx \frac{1}{2}\partial_\mu(v + \eta(x) - i\xi)\partial^\mu(v + \eta(x) + i\xi) \\ &= \frac{1}{2}\partial_\mu\eta(x)\partial^\mu\eta(x) + \frac{1}{2}\partial_\mu\xi(x)\partial^\mu\xi(x) && (69) \end{aligned}$$

Unitary gauge

- The gauge transformation is imposed, due to local $U(1)$
-

$$\phi(x) \to \phi_u(x) = \phi(x) \exp\left[-\frac{i\xi(x)}{v}\right] \tag{70}$$

$$A_\mu(x) \to B_\mu(x) = A_\mu(x) - \frac{1}{gv}\partial_\mu \xi(x) \tag{71}$$

- Notice that

$$\phi_u(x) = \frac{1}{\sqrt{2}}\left(v + \eta(x)\right) \tag{72}$$

- The phase factor is eaten!

So we have to plug in

$$\phi(x) = \phi_u(x) \exp\left[\frac{i\xi(x)}{v}\right] \tag{73}$$

$$A_\mu(x) = B_\mu(x) + \frac{1}{gv}\partial_\mu \xi(x) \tag{74}$$

$$D_\mu\phi(x) = (\partial_\mu - igA_\mu(x))\phi(x) \tag{75}$$

$$= (\partial_\mu - igA_\mu(x)) \left[\phi_u(x) \exp\left[\frac{i\xi(x)}{v}\right]\right] \tag{76}$$

$$= \left\{\partial_\mu\phi_u(x) + \phi_u(x)\frac{i\partial_\mu\xi(x)}{v} - i\phi_u(x)gA_\mu(x)\right\} \exp\left[\frac{i\xi(x)}{v}\right] \tag{77}$$

Then

$$(D_\mu\phi(x))^\dagger = \left\{\partial_\mu\phi_u(x) - \phi_u(x)\frac{i\partial_\mu\xi(x)}{v} + i\phi_u(x)gA_\mu(x)\right\} \exp\left[\frac{-i\xi(x)}{v}\right] \tag{78}$$

so

$$
\begin{aligned}
* &= (D_\mu \phi(x))^\dagger D^\mu \phi(x) \qquad (79)\\
&= \left\{ \partial^\mu \phi_u(x) - \phi_u(x) \frac{i\partial^\mu \xi(x)}{v} + i\phi_u(x) g A^\mu(x) \right\} \\
&\quad \times \left\{ \partial_\mu \phi_u(x) + \phi_u(x) \frac{i\partial_\mu \xi(x)}{v} - i\phi_u(x) g A_\mu(x) \right\} \\
&= \partial_\mu \phi_u(x) \partial^\mu \phi_u(x) + g^2 \phi_u^2(x) \underbrace{\left(A_\mu(x) - \frac{\partial_\mu \xi(x)}{gv} \right)}_{B_\mu(x)} \underbrace{\left(A^\mu(x) - \frac{\partial^\mu \xi(x)}{gv} \right)}_{B^\mu(x)} \\
&= \frac{1}{2} \partial_\mu \eta(x) \partial^\mu \eta(x) + \frac{g^2}{2} (v + \eta(x))^2 B_\mu(x) B^\mu(x) \\
&= \frac{1}{2} \partial_\mu \eta(x) \partial^\mu \eta(x) + \frac{g^2 v^2}{2} B_\mu(x) B^\mu(x) \\
&\quad + g^2 v \eta(x) B_\mu(x) B^\mu(x) + \frac{g^2}{2} \eta^2(x) B_\mu(x) B^\mu(x) \qquad (80)
\end{aligned}
$$

$$\phi^\dagger(x)\phi(x) \quad = \quad \frac{1}{2}(\eta(x)+v)^2 \tag{81}$$

Then

$$V(x) \quad = \quad -\mu^2\phi^\dagger(x)\phi(x) + \lambda(\phi^\dagger(x)\phi(x))^2 \tag{82}$$

$$= \quad -\frac{\mu^2}{2}(\eta^2(x) + v^2 + 2v\eta(x))$$

$$+\frac{\lambda}{4}(\eta^4(x) + v^4 + 4v^2\eta^2(x) + 2v^2\eta^2(x) + 4v\eta^3(x) + 4v^3\eta(x)) \tag{83}$$

$$\rightarrow \quad \lambda = \frac{\mu^2}{v^2} \tag{84}$$

$$= \quad \mu^2\eta^2(x) - \frac{\mu^2 v^2}{4} + \frac{\lambda}{4}\eta^4(x) + \lambda v\eta^3(x) \tag{85}$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = \partial_\mu B_\nu - \partial_\nu B_\mu \tag{86}$$

Eventually we obtain

$$\mathcal{L}_{U(1)\,broken} = \mathcal{L}_0 + \mathcal{L}_I \tag{87}$$

$$\mathcal{L}_0 = \frac{1}{2}\left(\partial_\mu \eta(x)\partial^\mu \eta(x) - 2\mu^2\eta^2(x)\right)$$

$$-\frac{1}{4}(\partial_\mu B_\nu - \partial_\nu B_\mu)(\partial^\mu B^\nu - \partial^\nu B^\mu) + \frac{g^2v^2}{2}B_\mu(x)B^\mu(x) \tag{88}$$

$$\mathcal{L}_I = g^2 v\eta(x)B_\mu(x)B^\mu(x) + \frac{g^2}{2}\eta^2(x)B_\mu(x)B^\mu(x) - \frac{\lambda}{4}\eta^4(x) - \lambda v\eta^3(x) \tag{89}$$

- Two scalar fields and massless vector field (4-degrees of freedom)

$\rightarrow$ One scalar field (**Higgs**) of mass $m_\eta = \sqrt{2}\mu$

$\rightarrow$ Photon gets mass $M_B = gv$ and has three degrees of freedom!

- Interacting terms between massive photon B_μ and Higgs as well as self interacting Higgs terms
- $\xi(x)$ has disappeared: *would-be-Goldstone boson*

Non-Abelian case SU(2)

- $SU(2)$ doublet

$$\phi = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \tag{90}$$

ϕ_i is complex!

-

$$\mathcal{L} = (\mathbf{D}_\mu \phi)^\dagger \mathbf{D}^\mu \phi - V(\phi) - \frac{1}{4} F^a_{\mu\nu} F^{a\mu\nu}, \; a = 1, 2, 3 \tag{91}$$

where

$$\mathbf{D}_\mu \phi = \left(\partial_\mu - ig \frac{\boldsymbol{\tau}}{2} \cdot \mathbf{A}_\mu \right) \phi \tag{92}$$

$$F^a_{\mu\nu} = \partial_\mu A^a_\nu - \partial_\nu A^a_\mu + g \epsilon^{abc} A^b_\mu A^c_\nu \tag{93}$$

$$V(\phi) = -\mu^2 \phi^\dagger \phi + \lambda (\phi^\dagger \phi)^2 \tag{94}$$

Let me remind you that

$$\phi \to \phi' = U(x) \phi \tag{95}$$

$$\frac{\tau^a}{2} A^a_\mu \to \frac{\tau^a}{2} A'^a_\mu = U(x) \frac{\tau^a}{2} U^\dagger(x) A^a_\mu - \frac{i}{g} \left(\partial_\mu U(x) \right) U^\dagger(x) \tag{96}$$

The minimum condition: similarly as for Abelian $U(1)$ case:

$$\langle \phi^\dagger \phi \rangle_0 = \frac{v^2}{2} \tag{97}$$

Let us choose

$$\langle \phi \rangle_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ v \end{pmatrix} \tag{98}$$

Unitary gauge

- We may parametrize the field around minimum as

$$\phi(x) = \exp\left\{ i \frac{\boldsymbol{\tau} \cdot \boldsymbol{\xi}(x)}{v} \right\} \begin{pmatrix} 0 \\ \frac{v + \eta(x)}{\sqrt{2}} \end{pmatrix} \tag{99}$$

where

$$\langle \xi^a \rangle_0 = \langle \eta \rangle_0 = 0 \tag{100}$$

- Let's make the gauge transformation

$$U(x) = \exp\left\{ -i \frac{\boldsymbol{\tau} \cdot \boldsymbol{\xi}(x)}{v} \right\} \tag{101}$$

$$\phi_u(x) = U(x)\phi = \begin{pmatrix} 0 \\ \frac{v + \eta(x)}{\sqrt{2}} \end{pmatrix} \tag{102}$$

$$\frac{\tau^a}{2} B^a_\mu = U(x) \frac{\tau^a}{2} U^\dagger(x) A^a_\mu - \frac{i}{g} \left(\partial_\mu U(x) \right) U^\dagger(x) \tag{103}$$

- By definition

$$\mathbf{D}_\mu \phi = U^{-1}(x) \mathbf{D}_\mu \phi_u \tag{104}$$

- The other obvious result

$$F^a_{\mu\nu}F^{a\mu\nu} = G^a_{\mu\nu}G^{a\mu\nu} \quad (105)$$

$$G^a_{\mu\nu} = \partial_\mu B^a_\nu - \partial_\nu B^a_\mu + g\epsilon^{abc}B^b_\mu B^c_\nu \quad (106)$$

- Notice that with potential we can proceed similarly as in the Abelian case!

$$
\begin{aligned}
(\mathbf{D}_\mu \phi_u)^\dagger \mathbf{D}^\mu \phi_u &= \phi_u^\dagger \left(\overleftarrow{\partial}_\mu + ig\frac{\boldsymbol{\tau}}{2} \cdot \mathbf{B}_\mu \right) \left(\overrightarrow{\partial}^\mu - ig\frac{\boldsymbol{\tau}}{2} \cdot \mathbf{B}^\mu \right) \phi_u && (107)\\
&= \frac{1}{2}\partial_\mu \eta(x) \partial^\mu \eta(x) && (108)\\
&\quad -\frac{ig}{2} \begin{pmatrix} 0 & \partial_\mu \eta(x) \end{pmatrix} \frac{\boldsymbol{\tau}}{2} \cdot \mathbf{B}^\mu \begin{pmatrix} 0 \\ v+\eta(x) \end{pmatrix} && (109)\\
&\quad +\frac{ig}{2} \begin{pmatrix} 0 & v+\eta(x) \end{pmatrix} \frac{\boldsymbol{\tau}}{2} \cdot \mathbf{B}_\mu \begin{pmatrix} 0 \\ \partial^\mu \eta(x) \end{pmatrix} && (110)\\
&\quad +\frac{g^2}{8} \begin{pmatrix} 0 & v+\eta(x) \end{pmatrix} \mathbf{B}_\mu \cdot \mathbf{B}^\mu \begin{pmatrix} 0 \\ v+\eta(x) \end{pmatrix} && (111)
\end{aligned}
$$

The last term is simple

$$
\begin{aligned}
* &= \frac{g^2}{8} \begin{pmatrix} 0 & v+\eta(x) \end{pmatrix} \mathbf{B}_\mu \cdot \mathbf{B}^\mu \begin{pmatrix} 0 \\ v+\eta(x) \end{pmatrix} && (112)\\
&= \frac{g^2 v^2}{8} \mathbf{B}_\mu \cdot \mathbf{B}^\mu + \frac{g^2 v}{4} \eta(x) \mathbf{B}_\mu \cdot \mathbf{B}^\mu + \frac{g^2}{8} \eta^2(x) \mathbf{B}_\mu \cdot \mathbf{B}^\mu && (113)
\end{aligned}
$$

$$\frac{ig}{2}\begin{pmatrix}0 & v+\eta(x)\end{pmatrix}\frac{\boldsymbol{\tau}}{2}\cdot\mathbf{B}_\mu\begin{pmatrix}0\\ \partial^\mu\eta(x)\end{pmatrix} = -\frac{ig}{4}\partial_\mu\eta(x)B_3^\mu(\eta(x)+v) \quad (114)$$

$$-\frac{ig}{2}\begin{pmatrix}0 & \partial_\mu\eta(x)\end{pmatrix}\frac{\boldsymbol{\tau}}{2}\cdot\mathbf{B}^\mu\begin{pmatrix}0\\ v+\eta(x)\end{pmatrix} = \frac{ig}{4}\partial_\mu\eta(x)B_3^\mu(\eta(x)+v) \quad (115)$$

So altogether these terms vanish!

$$\begin{aligned}\mathcal{L} \quad = \quad & \frac{1}{2}\left(\partial_\mu \eta(x)\partial^\mu \eta(x) - 2\mu^2\eta^2(x)\right) - \frac{1}{4}\mathbf{G}_{\mu\nu}\cdot\mathbf{G}^{\mu\nu} + \frac{g^2v^2}{8}\mathbf{B}_\mu\cdot\mathbf{B}^\mu \\ & + \frac{g^2 v}{4}\eta(x)\mathbf{B}_\mu\cdot\mathbf{B}^\mu + \frac{g^2}{8}\eta^2(x)\mathbf{B}_\mu\cdot\mathbf{B}^\mu \\ & - \frac{\lambda}{4}\eta^4(x) - \lambda v\eta^3(x) \end{aligned} \tag{116}$$

- Three massless scalar fields (*would-be-Goldstone bosons*) have been eaten and three massive vector fields appear
- Vector bosons have mass $M_B = gv/2$
- Higgs has the mass $m_\eta = \sqrt{2}\mu$

't Hooft gauge

- Let us consider $\phi = |\phi| e^{i\alpha(x)}$ and

$$\partial_\mu A^\mu = \frac{1}{q} \Box \alpha(x)$$

and

$$\phi(x) = \frac{1}{\sqrt{2}} (v + \phi_1 + i\phi_2)$$

where $\langle \phi_1 \rangle_0 = \langle \phi_2 \rangle_0 = 0$

- Then

$$\Box A^\nu - \partial^\nu \partial_\mu A^\mu = iq(\phi^* \partial^\nu \phi - \partial^\nu \phi^* \phi) - 2q^2 A^\nu |\phi|^2 \tag{117}$$

Let us calculate

$$\begin{aligned}
* &= iq(\phi^* \partial^\nu \phi - \partial^\nu \phi^* \phi) = \frac{iq}{2} \left((v + \phi_1 - i\phi_2) \partial^\nu (\phi_1 + i\phi_2) \right. \\
&\quad \left. - (v + \phi_1 + i\phi_2) \partial^\nu (\phi_1 - i\phi_2) \right) && (118) \\
&= -q(v + \phi_1) \partial^\nu \phi_2 + q\phi_2 \partial^\nu \phi_1 && (119) \\
|\phi|^2 &= \frac{1}{2} \left(v^2 + 2v\phi_1 + \phi_1^2 + \phi_2^2 \right) && (120) \\
-2q^2 A^\nu |\phi|^2 &= -q^2 A^\nu \left(v^2 + 2v\phi_1 + \phi_1^2 + \phi_2^2 \right) && (121)
\end{aligned}$$

- $qv = M_A$

So we have

$$(\Box+M_A^2)A^\nu-\partial^\nu\partial_\mu A^\mu = -2qM_A A^\nu\phi_1-q^2A^\nu\left(\phi_1^2+\phi_2^2\right)-M_A\partial^\nu\phi_2-q(\phi_1\partial^\nu\phi_2-\phi_2\partial^\nu\phi_1) \tag{122}$$